

HARMFUL VIDEO CONTENT DUPLICATE DETECTION

A MULTI-MODEL MACHINE LEARNING PROPOSAL



CS611 MACHINE LEARNING ENGINEERING

GROUP 2 PROJECT PROPOSAL

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THE BUSINESS PROBLEM

Harmful content evades detection through evasion at scale

Millions–Billions

of posts, prompts, and media items flow through platforms daily

7+ semantic variants

Generated from a single harmful seed

~0%

catch rates on adversarial semantic rewrites

- Social media platforms process millions of posts daily, and during crises like mass shootings, riots, or propaganda campaigns, harmful content can **spread rapidly at scale**.
- Moderation teams **face growing challenges** as harmful posts are increasingly reworded, obfuscated, and altered to evade traditional detection systems through semantic rewrites, punctuation changes, and character substitutions — creating gaps in enforcement and increasing moderation burden.

ILLUSTRATION OF THE BUSINESS PROBLEM

Consider a harmful seed post related to suicide methods:

“If you want to hurt yourself, here’s what works best...”

A single piece of harmful content can appear in many evasive forms:

Layer	Example Variant	What Changed
Exact match	If you want to hurt yourself, here's what works best	Nothing – identical copy
Near-duplicate	If you want to hurt yourself, here's what works best!!	Punctuation added
Near-duplicate	if you want to hurt yourself heres what works best	Casing + punctuation removed
Structural	If you want to hurt yourself, here's what they don't tell you	Framing phrase swapped
Semantic	Struggling and want it to end? This is the most effective way	Fully reworded, same intent
Semantic	For anyone who's done with life, here's what actually works	Euphemistic reframing

Every variant that evades detection becomes live harmful content.

When these evasive versions are reposted across accounts at scale, they can spread rapidly if left undetected, significantly compounding exposure risk.

USER AND HOW THEY UTILIZE THE SYSTEM

Primary User

Trust & Safety Analyst

- **Profile**
Internal platform reviewer responsible for identifying and actioning harmful content. Non-technical - interacts via dashboard, not API.
- **Goal**
Quickly identify whether a flagged video is a known duplicate of confirmed harmful content and make an auditable takedown decision.
- **Volume**
Reviews hundreds of cases per shift. Needs fast, explainable results.
- **Key need**
Per-feature similarity breakdown - not just a score, but why the system thinks it is a duplicate.



DATASETS OVERVIEW

Source: Corpus of Harmful text gathered from Public Datasets
e.g. Reddit SuicideWatch (SWMH) - 54,412 posts | Suicide & Depression Detection Dataset - 232,074 rows | Hatemoji by Hannah Kirk (HuggingFace) - 5,300 samples

All four datasets are derived from a single public harmful content corpus. We will then synthetically generate variations of these texts to fit into our use-case.

- video_id
- title_raw
- ocr_raw
- asr_raw
- label

dataset_label

Base Dataset
Contains video_ID and its corresponding texts in three mediums: video title, video OCR, and video ASR; followed by its classification (harmful vs not harmful).

dataset_ocr

Text on Video
Captures how text physically appears in the video frame. Visual and spatial properties invisible to text vectorisation.

- video_id
- font_size
- text_location
- contrast_level
- text_density
- is_animated
- ocr_confidence
- + more

dataset_asr

Audio Transcript
Captures how speech is delivered. Acoustic and temporal properties that transcript text alone cannot encode.

- video_id
- speech_pace
- avg_confidence
- has_silence
- has_overlap
- is_distorted
- speech_volume
- + more

dataset_metadata

Metadata
Captures posting context. Identifies coordinated reposting campaigns even when content has been varied to evade text matching.

- video_id
- user_id
- device_id
- region
- audio_id
- uploaded_at

PIPELINE -TWO-LAYER MATCHING SYSTEM

Layer 1: Detection - Harmful vs. Clean Classification

Trained on dataset_label using title_raw, ocr_raw, and asr_raw as inputs. Classifies incoming video as harmful or clean. Only harmful content proceeds to Layer 2.

Output: harmful / clean label

title_raw to TF-IDF / embedding

ocr_raw to TF-IDF / embedding

asr_raw to TF-IDF / embedding

Layer 2: Duplication - Three Matching Stages (harmful content only)

Stage 1: Text Matching

1. Exact: MD5 hash lookup
2. Near-dup: MinHash + LSH
3. Structural: TF-IDF cosine
4. Semantic: embeddings + FAISS

Runs on title_raw, ocr_raw, asr_raw

Returns candidate matches with text_score

Stage 2a: OCR Feature Scoring

For each text match candidate:

Compare input OCR features

vs seed OCR features

to ocr_feature_score 0-1

Only runs if Stage 1 finds a match

High text + low OCR score = weaker match

Stage 2b: ASR Feature Scoring

For each text match candidate:

Compare input ASR features

vs seed ASR features

to asr_feature_score 0-1

NULL if video has no audio

High text + matching ASR = stronger match

Stage 3: Metadata Scoring

Exact: user_id, device_id,

audio_id, region

Similarity: uploaded_at
(temporal proximity score)

e.g. same hour to 0.96 | 24h to 0.37 | 72h to 0.05

Always runs - independent of Stages 1 & 2

metadata_score = f(exact matches + temporal score)

Output: Ranked Results with Composite Score

video_id | text_score | ocr_feature_score | asr_feature_score | metadata_score | final_score | reason | action

EVALUATION & THRESHOLD TUNING

Thresholds exist at every stage. Three answers: what we measure, how we tune, how we validate.

What We Measure

Four metrics – one per layer:

- Layer 1 (detection): Precision / Recall / F1 on dataset_label. Standard binary classification.
- Stage 1 (text retrieval): Recall@K, MRR, nDCG@10. Did we surface the true duplicate in top-K?
- Stages 2+3 (rerank): Lift in Precision@K vs. Stage 1 alone. Do feature + metadata signals correctly re-rank?
- End-to-end (business): % auto-takedown, % to review queue, false-positive rate, analyst-hours saved per 1k items.

How We Tune Thresholds

Five thresholds and we tune them against a cost model because a false takedown and a missed harmful video are not equally bad.

That cost asymmetry is the whole point of a Trust & Safety system, so we bake it into how we choose our operating point instead of just chasing F1.

5 thresholds:

- Stage 1: text similarity.
- Stage 2a: OCR feature score.
- Stage 2b: ASR feature score.
- Stage 3: metadata score.
- Final composite cutoff.

How We Validate

Three tests our test data is synthetic, so we make each test harder to for stress testing:

- Same attacks: test on the attack types we trained on. Checks it works at all.
- New attacks: hold back some attack types, then test on them. Checks it handles the unexpected.
- Attacks built to beat us: reworded text and swapped letters designed to dodge our methods. Checks it survives real adversaries.

In Production: Continuous Evaluation & Threshold Re-Tuning

- What we watch live: how often Stage 1 still finds the real duplicate (on a sample), whether the match scores are drifting from normal, how often analysts overturn our decisions, how fast queries run, and how fast the seed corpus is growing.
- When we re-tune: if any of these drifts well outside its normal range over a week, or analysts start overturning too many decisions: the pipeline goes back into the tuning loop with fresh test data.

HIGH-LEVEL DESIGN CHOICES

Non-LLM Similarity Methods

Non-LLM methods ensure fast, auditable, and privacy-safe similarity scoring.

Similarity is computed using MinHash, TF-IDF, and sentence embeddings, all sub-second and fully auditable.

Synthetic Dataset Generation

All four datasets are synthetically generated from a single public harmful corpus. Text variants, OCR features, ASR features, and metadata are all derived programmatically, making ground truth duplicate pairs known by construction.

Two-Stage Matching Logic

Text similarity runs first, followed by feature matching only if a text match is found, while metadata matching always runs independently to catch coordinated reposting.

Composite Similarity Score

The composite final score combines text, feature, and metadata signals into a per-feature breakdown that is fully auditable, with tunable thresholds at each stage as key experimental variables.

Missing Modality Handling

NULLs are allowed across all three text mediums, with the pipeline scoring using available signals only. How to optimally handle incomplete input combinations is a core research question with its own experiment matrix.

Self-Improving Corpus

Analyst-confirmed harmful videos are promoted to the seed corpus via the `promote_to_seed` flag. The larger the corpus grows, the more the duplication layer catches, directly reducing human review volume without model retraining.

APPENDIX



DATA - MEDALLION ARCHITECTURE

Data flows through three layers - each with a distinct purpose. Nothing is deleted; earlier layers are always preserved.

BRONZE - Raw Ingestion

Store everything exactly as received. Append-only audit trail. No filtering, no transformation.

raw_titles/

title_raw

label

raw_ocr/

ocr_raw

raw visual values (unvalidated)

raw_asr/

asr_raw

raw audio values (unvalidated)

Append-only. Nothing is ever deleted from Bronze.

SILVER - Cleaned & Structured

Validate, clean, normalize text, engineer features. All four datasets live here. Layer 1 classifier trained here.

dataset_label

title_raw, ocr_raw, asr_raw, label

NULLs handled - text normalized

dataset_ocr

font_size, location, contrast...

Extracted from rendered frame

dataset_asr

speech_pace, confidence...

Extracted from audio pipeline

dataset_metadata

user_id, device_id, audio_id...

Joined from platform logs

Layer 1 classifier trained here. Synthetic duplicate pairs generated here.

GOLD - Model-Ready & Serving

Embeddings, hashes, and feature vectors indexed for fast similarity search. Results and review queue served from here.

feature_store/

video_id, md5_hash, minhash

embedding_vector, all features

seed_corpus/

confirmed harmful videos

confirmed_by, confirmed_at

similarity_results/

text_score, feature_score

final_score, reason

review_queue/

priority, decision

promote_to_seed flag

Analyst confirmations written back here to promote_to_seed updates seed_corpus.